Skills Summary

Two-Way Tables: Comparing Categorical Frequencies

Use this reference while completing your worksheet or when studying.

Every example below reads this table. A vet clinic recorded 100 pets by species and by where they live:

	Indoor	**Outdoor**	**Row total**
Cats	21	9	30
Dogs	14	56	70
Column total	35	65	100

Skill 1 — Reading a two-way table

Rule: a cell is where one row meets one column, so it counts the individuals in *both* categories at once. A **row total** is that row's cells added up; a **column total** is that column's cells added up; the **grand total** is every inner cell added up, and it must equal both the sum of the row totals and the sum of the column totals.

Example: How many pets are dogs that live outdoors?
Step 1. "Dogs" and "outdoors" are two categories at once, so the answer is a single cell — find the row, then the column.
Step 2. Read across the Dogs row and down the Outdoor column; they meet at one number.
56 dogs live outdoors.

Try it: How many pets are cats that live outdoors?

check: **9**

Skill 2 — Row relative frequency

Rule: divide a cell by *its own row* total, then multiply by 100. Row relative frequency answers "of the ones in this row, what percent are in this column?" The percents along one row always add to 100.

Example: Of the cats, what percent live indoors?
Step 1. The question starts with "of the cats," so the group being described is the Cats row — that row's total is the base.
Step 2. The cell is 21 and the Cats row total is 30, so $100 \times \frac{21}{30} = 100 \times 0.7$.
70% of the cats live indoors.

Try it: Of the dogs, what percent live indoors?

check: **20%**

Skill 3 — Column relative frequency

Rule: divide the same kind of cell by *its own column* total, then multiply by 100. Column relative frequency answers the reversed question: "of the ones in this column, what percent are in this row?" Same cell, different base, different answer.

Example: Of the indoor pets, what percent are cats?
Step 1. This time the group is "the indoor pets," which is a column, so the Indoor column total is the base — not the Cats row total.
Step 2. The cell is 21 and the Indoor column total is 35, so $100 \times \frac{21}{35} = 100 \times 0.6$.
60% of the indoor pets are cats.

Try it: Of the indoor pets, what percent are dogs?

check: **40%**

Skill 4 — Comparing to judge association

Rule: compute the same relative frequency for both rows, then subtract to get the gap in **percentage points**. A large gap means the two variables are **associated** — knowing one changes what you expect of the other. A gap near zero means little or no association. Never compare raw counts across rows of different sizes.

Example: 70 percent of cats live indoors and 20 percent of dogs do. Is species associated with where the pet lives?
Step 1. The two rows hold different numbers of pets, so compare the two row percents, not the counts 21 and 14.
Step 2. The gap is $70 - 20$ percentage points, which is large.
50 percentage points — yes, there is an association.

Try it: In another clinic 45 percent of cats and 42 percent of dogs live indoors. Find the gap, then say whether there is an association.

check: **3** points — no real association

(!) Watch out: The most costly mistake on this topic is using the wrong total. The same cell can give three different percents — out of its row, out of its column, or out of the grand total — and they answer three different questions. Before you divide, read the phrase "of the ..." in the question: whatever follows it names the total you need.